

11 - K-Means Clustering

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Explain unsupervised clustering and centroid-based partitioning.
- Implement the K-Means update loop.
- Use inertia and the elbow method as practical diagnostics.

Background

K-Means partitions unlabeled numerical data into k groups. The method alternates between assigning examples to centroids and updating centroids. Different initializations can produce different local solutions, so multiple runs are common.

Core Concepts

Concept	Meaning in this lab
Cluster	Group of examples treated as similar.
Centroid	Mean feature vector of a cluster.
Assignment step	Assign each example to its nearest centroid.
Update step	Recompute each centroid from assigned examples.
Inertia	Sum of squared distances from points to their assigned centroids.

Guided Procedure

1. Select two or more numerical features and standardize them when scales differ substantially.
2. Initialize k centroids from data points.
3. Assign every point to the nearest centroid.
4. Update each centroid to the mean of its assigned points.
5. Repeat until assignments stop changing or movement is small.
6. Run multiple k values and inspect inertia.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
import numpy as np

def kmeans_step(X, centroids):
    d = ((X[:, None, :] - centroids[None, :, :]) ** 2).sum(axis=2)
    labels = d.argmin(axis=1)
    new_centroids = np.vstack([
        X[labels == j].mean(axis=0) for j in range(len(centroids))
    ])
    return labels, new_centroids
```

Lab Exercises

- Task 1.** Implement the complete loop and stop when centroid movement is below a threshold.
- Task 2.** Compare $k=2$ through $k=8$ using inertia.
- Task 3.** Visualize a two-feature dataset with points colored by cluster.

Task 4. Explain what should happen if a cluster receives no assigned points and implement a remedy.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. Why is K-Means unsupervised?
2. Why can initialization affect the result?
3. What does the elbow method attempt to balance?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.